Knowledge Distillation Across Teacher Architectures for Edge-Efficient Plant Disease Classification

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Abstract. Accurate image-based plant disease classification supports precision agriculture, but high-capacity networks are inconvenient for edge or mobile deployment. This work studies knowledge distillation from large teachers into a compact MobileNetV3-Small student, asking in particular whether the *architecture of the teacher* affects the distilled student. As a preliminary demonstration on the 38-class PlantVillage dataset (single split), a ResNet50 teacher distills into a student reaching 0.9937 accuracy and 0.9907 macro-F1 with 1.56M parameters, about 15 times fewer than the teacher. The main study uses the smaller, class-imbalanced Cassava leaf-disease dataset (five classes) under stratified five-fold cross-validation: three teachers from different architectural families — ResNet50 (residual CNN), EfficientNet-B2 (compound-scaled CNN), and ViT-B/16 (vision transformer) — are each retrained per fold and distilled into the same student. Distillation improves the student over a supervised baseline by 1.4–1.8 accuracy points, but the choice of teacher has little effect: the three students agree to within 0.5 points despite the teachers differing widely in accuracy (82.4–85.1%) and size (7.7M–85.8M parameters). The benefit of distillation thus appears to come from the procedure itself rather than from the capacity or architecture of the teacher.

Keywords: Knowledge distillation · Plant disease classification · MobileNetV3 · Cross-validation · Vision transformer · Edge inference.

1 Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture. Deep neural networks can achieve strong classification accuracy on curated leaf image datasets, but models with high parameter counts may be inconvenient for mobile or edge-oriented deployment, creating a tension between predictive accuracy and computational efficiency.

This work studies that trade-off through knowledge distillation, in which a high-capacity teacher network guides the training of a lightweight MobileNetV3-Small student. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain a teacher's predictive behavior while using far fewer parameters. Beyond the standard teacher-to-student comparison, we focus on a question that a single experiment cannot answer: does the *choice of teacher architecture* change how well the student learns?

We make three contributions: (i) a preliminary single-split study on 38-class PlantVillage confirming that distillation improves a compact student; (ii) a cross-validated study on Cassava comparing three teacher architectures (ResNet50, EfficientNet-B2, ViT-B/16) under stratified five-fold cross-validation, with the teacher retrained per fold and paired significance tests against the baseline; and (iii) a post-training quantization analysis of the distilled student.

2 Related Work

PlantVillage [8] provides a widely used benchmark of labeled plant leaf images. Its clean imaging conditions are well suited to initial controlled experiments but less representative of field conditions. The Cassava leaf-disease dataset [12] covers five classes with strong class imbalance and real field imagery, providing a harder and more realistic testbed. Early work demonstrated high accuracy on PlantVillage with convolutional networks [10], and later studies compared fine-tuning strategies across architectures [14], but the literature has paid comparatively little attention to compressing these models for on-device inference.

Knowledge distillation [6] transfers information from a larger teacher to a smaller student by having the student match the teacher's softened output probabilities. Numerous extensions go beyond the output logits — for example, matching intermediate feature maps (FitNets [11]) or spatial attention maps [16] — and a broad survey is given by Gou et al. [4]. We use the original logit-based formulation since our focus is the teacher architecture rather than the distillation mechanism.

A recurring finding is that stronger teachers do not necessarily produce better students. Cho and Hariharan [2] show that a large capacity gap can impair distillation, and Mirzadeh et al. [9] propose intermediate teacher assistants to bridge it. Beyer et al. [1] argue that training consistency matters more than teacher accuracy. These results motivate our central question: when the student is fixed, does the teacher's *architecture* — and not merely its accuracy — affect the distilled student? DeiT [15] distills *into* a transformer using a CNN teacher; our setting is complementary, distilling *from* a transformer and two CNN teachers into a convolutional student.

The three teachers span distinct architectural families. ResNet50 uses residual connections to ease the optimization of deeper convolutional models [5]; EfficientNet-B2 applies compound scaling of depth, width, and resolution [13]; and ViT-B/16 treats an image as a sequence of patches processed by self-attention [3]. The student in every experiment is MobileNetV3-Small [7], a compact network designed for mobile inference.

3 Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. Supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen, and the student is optimized using a weighted sum of the cross-entropy loss and a temperature-scaled Kullback–Leibler divergence term:

$$\mathcal{L} = \alpha T^2 \mathrm{KL}\Big(\mathrm{softmax}\Big(\frac{z_t}{T}\Big) \Big\| \mathrm{softmax}\Big(\frac{z_s}{T}\Big)\Big) + (1-\alpha)\,\mathrm{CE}(y, z_s). \tag{1}$$

The T^2 factor compensates for the gradient scaling introduced by softened probabilities. The temperature is fixed at $T = 4.0$ throughout; the distillation weight is $\alpha = 0.5$ in the PlantVillage study and $\alpha = 0.7$ in the main Cassava study. In both cases α is held constant across teachers so that any difference between teachers is attributable to the teacher rather than to the loss weighting.

4 Experimental Setup

All experiments use pretrained ImageNet weights, images resized to 224×224 pixels and normalized with ImageNet statistics, random horizontal flipping and small rotations during training, deterministic resizing for validation and test, and the best validation checkpoint for evaluation. Latency is measured as a batch-one average over repeated forward passes on local hardware and should not be read as a deployment benchmark.

4.1 Main Study: Cassava Cross-Validation

The main study uses the Cassava leaf-disease dataset (five classes) under stratified five-fold cross-validation. For each fold, the teacher is retrained from scratch on that fold's training partition and then frozen before distilling the student; this avoids the bias that would arise if a single teacher had seen images that later appear in the test fold. The fold partition is fixed by seed, so every model sees identical folds. Three teachers are evaluated — ResNet50, EfficientNet-B2, and ViT-B/16 — each distilled into MobileNetV3-Small and compared against a supervised baseline; per-fold training schedules are given in Table 1. Results are reported as mean ± standard deviation across the five folds, and each distilled student is compared against the baseline with a paired two-sided t-test on per-fold accuracies. All runs use an NVIDIA RTX 4070 with automatic mixed precision.

4.2 Preliminary Study: PlantVillage Single Split

As a preliminary demonstration, a ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled student are trained on the full 38-class PlantVillage

Table 1. Per-fold training configuration for the Cassava cross-validation study. All teachers are distilled into MobileNetV3-Small with $T = 4.0$ and $\alpha = 0.7$.

Model	Role	Learning rate	Epochs	Batch
ResNet50	Teacher	3×10^{-4}	10	64
EfficientNet-B2	Teacher	3×10^{-4}	10	64
ViT-B/16	Teacher	1×10^{-4}	10	32
MobileNetV3-Small	Student / baseline	5×10^{-4}	15	64

dataset with a single 70/15/15 train/validation/test split (seed 42), 15 epochs, batch size 32, $T = 4.0$, $\alpha = 0.5$, on Apple Silicon (MPS). Learning rates are 3×10^{-4} (teacher), 10^{-3} (supervised student), and 5×10^{-4} (distilled student).

5 Results

Experiments cover two datasets. On PlantVillage (38 classes, single split), we train one ResNet50 teacher, one supervised MobileNetV3-Small baseline, and one distilled student. On Cassava (five classes, five-fold CV), we train three teacher baselines, one student baseline, and three distilled students — one per teacher — plus an ablation over distillation hyperparameters and two post-training quantization variants. All metrics are macro-averaged precision, recall, and F1 alongside accuracy.

5.1 Cassava Cross-Validation

Table 2 summarizes the cross-validated results.

Table 2. Cross-validated test results on cassava (5-fold, mean ± standard deviation across folds). Precision, recall, and F1 are macro-averaged. The teacher is retrained per fold; the student and KD hyperparameters are fixed.

Teacher	Student	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
–	resnet50	85.13 ± 0.95	75.08 ± 1.40	73.69 ± 1.28	74.00 ± 0.78	23,518,277
–	efficientnet_b2	84.59 ± 0.67	73.91 ± 1.51	71.75 ± 0.86	72.60 ± 0.61	7,708,039
resnet50	mobilenet_v3_small	83.49 ± 0.31	72.70 ± 0.66	69.74 ± 2.62	70.72 ± 1.99	1,522,981
vit_b_16	mobilenet_v3_small	83.12 ± 0.72	72.90 ± 0.99	68.25 ± 2.36	69.85 ± 1.95	1,522,981
efficientnet_b2	mobilenet_v3_small	83.04 ± 0.41	71.75 ± 1.53	69.40 ± 1.98	70.34 ± 1.26	1,522,981
–	vit_b_16	82.40 ± 0.64	71.41 ± 1.28	69.08 ± 1.58	69.38 ± 1.26	85,802,501
–	mobilenet_v3_small	81.67 ± 1.01	69.83 ± 2.64	67.57 ± 1.38	68.22 ± 0.80	1,522,981

Distillation helps. All three teachers raise the student above the supervised baseline (81.67% accuracy) by a similar margin: +1.82 points for ResNet50 (83.49%), +1.45 for ViT-B/16 (83.12%), and +1.37 for EfficientNet-B2 (83.04%). In the paired t-test the ResNet50 and ViT-B/16 gains are statistically significant

($p = 0.029$ and $p = 0.036$); the EfficientNet-B2 gain does not reach significance ($p = 0.079$), most likely reflecting the limited power of five folds and that run's larger fold-to-fold variance.

The more notable result is how little the teacher matters. The three distilled students lie within 0.5 points of one another (83.04–83.49%), a spread no larger than their across-fold standard deviations, even though the teachers differ substantially in accuracy (82.40–85.13%), architectural family (two CNNs and one transformer), and size (7.7M, 23.5M, 85.8M parameters). The ViT-B/16 student (83.12%) surpasses its own teacher (82.40%), which runs counter to the expectation that a very large teacher, separated from a 1.5M- parameter student by a wide capacity gap, would distill poorly. We interpret this teacher-agnostic behavior in Section 7.

5.2 Sensitivity to Distillation Hyperparameters

The main study fixes $T = 4.0$ and $\alpha = 0.7$ across teachers. To verify this choice is not an artifact, we run a single-split sensitivity analysis for the ResNet50 teacher on fold 0, sweeping $T \in \{1, 2, 4, 8\}$ at $\alpha = 0.7$ and $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ at $T = 4.0$.

Table 3. Single-split KD sensitivity analysis for the resnet50 teacher (fold 0 of the Cassava cross-validation split). Temperature is swept at $\alpha = 0.7$ and α at $T = 4$; † marks the default used in the main study. Precision, recall, and F1 are macro-averaged. Values are single-split point estimates (no fold error bars).

Sweep	T	α	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)
temperature	1	0.7	83.50	73.04	70.05	71.35
temperature	2	0.7	83.79	74.38	69.01	71.29
temperature	4	$0.7^{\dagger}$	84.14	73.99	70.29	71.52
temperature	8	0.7	83.67	73.26	70.03	71.37
alpha	4	0.3	82.78	70.59	70.76	70.61
alpha	4	0.5	83.60	72.91	69.64	70.89
alpha	4	$0.7^{\dagger}$	84.14	73.99	70.29	71.52
alpha	4	0.9	83.11	73.01	67.04	69.39

The student is stable: accuracy stays within one point over all settings (82.78–84.14%), with a gentle maximum at the default $T = 4.0$, $\alpha = 0.7$. Setting $\alpha = 0.9$ lowers macro-F1 to 69.39%, consistent with over-weighting the teacher at the expense of minority classes. The flat response confirms that the fixed configuration is well justified.

5.3 Edge Compression: Quantization

We apply two int8 post-training quantization (PTQ) strategies to the single-split distilled student on CPU using the `fbgemm` backend: *dynamic* PTQ (weights only) and *static* PTQ (weights and activations, calibrated on 256 training images). Table 4 reports the accuracy/size/latency trade-off.

Table 4. Post-training quantization of the distilled MobileNetV3-Small on Cassava (single-split test set, CPU, `fbgemm`). Precision, recall, and F1 are macro-averaged.

Variant	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Size (MB)	Latency (ms)
Float32 (baseline)	84.23	74.31	70.42	71.87	5.93	17.7
Dynamic int8	84.20	74.32	70.35	71.83	4.23	16.0
Static int8	12.22	21.06	20.43	6.05	1.74	18.4

Dynamic quantization is essentially lossless (84.23 $\rightarrow$ 84.20%, a 0.03-point change) and provides a free 1.4$\times$ size reduction. Static quantization reaches 3.4$\times$ compression but collapses accuracy to 12.2%, well below the 20% random baseline. This collapse reflects the known sensitivity of MobileNetV3-Small's depthwise-separable convolutions and Hardswish activations to post-training quantization; quantization-aware training would be required to recover accuracy at full int8 compression, which we leave to future work.

5.4 PlantVillage Single Split

Table 5 reports the single-split results on the 38-class PlantVillage task. The improvement from 0.9851 to 0.9937 accuracy after distillation shows that the teacher logits provide useful signal beyond the hard labels, even when the supervised student is already strong. This limited headroom motivates the main cross-validated study on the harder Cassava data.

Table 5. Single-split test results on the 38-class PlantVillage task. Precision, recall, and F1 are macro-averaged.

Model	Training strategy	Accuracy	Precision	Recall	Macro-F1	Parameters	Latency
ResNet50	Teacher supervised	0.9936	0.9920	0.9908	0.9912	23,585,894	5.47 ms
MobileNetV3-Small	Supervised baseline	0.9851	0.9788	0.9781	0.9779	1,556,806	3.51 ms
MobileNetV3-Small	Distilled from ResNet50	0.9937	0.9911	0.9907	0.9907	1,556,806	3.53 ms

6 Error Analysis

The gap between accuracy and macro-F1 throughout the Cassava results is a direct consequence of class imbalance. On the single-split test set, *mosaic disease* accounts for 1,984 of 3,209 images (62%), whereas *bacterial blight* has only 155 (5%). Table 6 and Figure 1 give the per-class performance and confusion matrix of the distilled student.

Table 6. Per-class performance of the distilled MobileNetV3-Small on the Cassava single-split test set.

Class	Precision	Recall	F1	Support
Bacterial blight	0.667	0.465	0.548	155
Brown streak disease	0.793	0.697	0.742	307
Green mottle	0.707	0.701	0.704	378
Healthy	0.616	0.712	0.660	385
Mosaic disease	0.934	0.947	0.940	1,984

The student performs well on the dominant class (mosaic disease, F1 0.94) but degrades on minority classes, reaching its worst on bacterial blight (F1 0.55, recall 0.47). Because accuracy is dominated by the majority class while macro-F1 weights all classes equally, accuracy (84%) substantially overstates per-class competence (macro-F1 72%), which is why we report both throughout. The confusion matrix reveals two systematic error patterns: minority diseases are frequently predicted as *healthy* (59 of 155 bacterial-blight leaves, 38%), and the visually similar *green mottle* and *mosaic disease* classes are mutually confused. Addressing these errors would likely require class re-balancing or cost-sensitive training.

A t-SNE projection of the student's penultimate embeddings (Figure 2) tells the same story geometrically: *mosaic disease* forms a large, well-separated cluster, consistent with its 0.94 F1, whereas *bacterial blight*, *green mottle*, and *healthy* overlap in a shared region — precisely the minority classes the confusion matrix shows being conflated. The representation is therefore well organized for the majority class but only weakly discriminative for the rare ones.

7 Discussion

The central observation is that the distilled student is largely insensitive to its teacher: across a residual CNN, a compound-scaled CNN, and a vision transformer — differing widely in accuracy, architecture, and size — the three students fall within 0.5 points of one another, and the weakest and largest teacher (ViT-B/16) distills as effectively as the strongest. This points to the softened-label regularization of distillation, rather than the teacher's capacity or architecture, as the source of the 1.4–1.8-point gain. The implication for edge settings is convenient: a small, inexpensive teacher suffices, and dynamic int8 quantization adds an orthogonal 1.4× size reduction at no accuracy cost.

Several limitations qualify these conclusions. Distillation hyperparameters are held fixed across teachers, so a teacher-specific T or α might widen the currently negligible gaps; with only five folds, statistical power is limited, as the borderline EfficientNet-B2 result illustrates. Both datasets are curated, so accuracy on field images with cluttered backgrounds and varying lighting may be lower, and latency is measured on local hardware rather than on a target edge device.

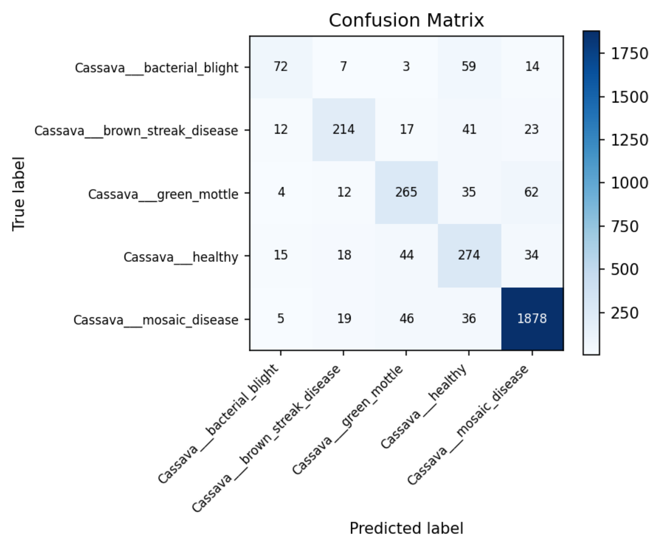

Fig. 1. Confusion matrix of the distilled MobileNetV3-Small on the Cassava single-split test set. Errors concentrate in the minority classes.

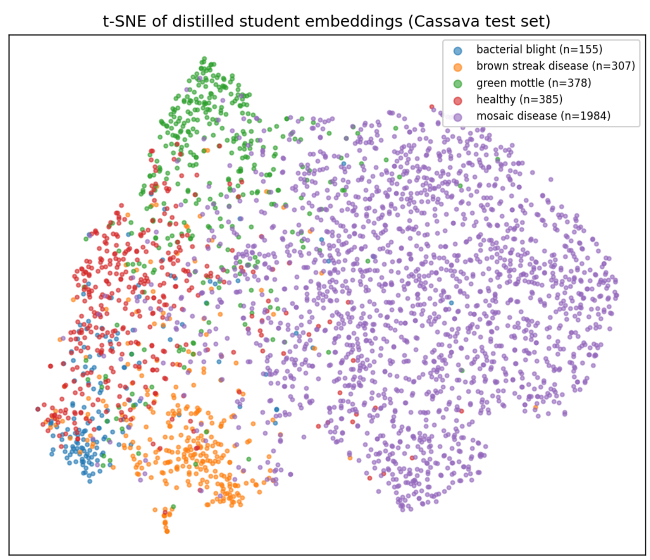

Fig. 2. t-SNE projection of the distilled student's 1024-dimensional penultimate embeddings on the Cassava single-split test set. The majority *mosaic disease* class separates cleanly, while the minority classes overlap, mirroring the per-class scores (Table 6) and the confusion matrix (Figure 1).

8 Conclusion

We studied knowledge distillation from large teachers into a compact MobileNetV3-Small student for edge-efficient plant disease classification. On PlantVillage, a ResNet50 teacher distilled into a student reaching 0.9937 accuracy with about 15× fewer parameters. The main cross-validated study on Cassava showed that distillation raises the student by 1.4–1.8 accuracy points over a supervised baseline, yet the choice of teacher — ResNet50, EfficientNet-B2, or ViT-B/16 — had little effect. The benefit therefore appears to stem from the distillation procedure itself rather than from the teacher, which is encouraging for deployment: a small, cheap teacher suffices, and dynamic int8 quantization compresses the student further at no accuracy cost. Full int8 compression, however, requires quantization-aware training to avoid an accuracy collapse.

Disclosure of Interests. The authors have no competing interests to declare.

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